

Lecture 05 — Binomial Trees, Finite Differences, and the Black–Scholes PDE

Computational Finance (WI4151), TU Delft — Prof. Fang Fang

Scope. This note covers the binomial (CRR) tree and its convergence to Black–Scholes; the finite-difference discretisation of the heat equation (FTCS, BTCS, Crank–Nicolson) with stencils, update equations, local truncation error and von Neumann stability; the application of these schemes to the Black–Scholes PDE; the reduction of the BS PDE to the heat equation; and the identification of the binomial method as a particular explicit finite-difference scheme. The cluster sits inside Cluster 1 (lattice / PDE methods). Index-based algebra is used throughout.

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1 The binomial method

1.1 Discrete asset model

Fix a maturity T and a number of steps M . Set the time increment $\delta t = T/M$ and the time nodes $t_i = i\delta t$, $0 \leq i \leq M$. Between two consecutive time slices the asset is assumed to move multiplicatively: it goes *up* by a factor u with probability p , or *down* by a factor d with probability $1 - p$. The tree is taken to be *recombining* (an up-then-down move lands at the same node as

down-then-up), so at time step i there are exactly $i + 1$ distinct nodes. Labelling by the number n of up-moves so far,

$$S_n^i = d^{i-n} u^n S_0, \quad 0 \leq n \leq i. \quad (1)$$

A recombining tree has $O(M^2)$ nodes rather than $O(2^M)$, which is what makes the method computationally cheap.

1.2 Risk-neutral backward induction

In an arbitrage-free market the option value is the discounted expected payoff under the risk-neutral measure, and this principle holds slice by slice in the tree. Writing $\Lambda(\cdot)$ for the payoff and V_n^i for the option value at node (i, n) :

$$V_n^M = \Lambda(S_n^M), \quad 0 \leq n \leq M, \quad (2)$$

$$V_n^i = e^{-r\delta t} \left(p V_{n+1}^{i+1} + (1-p) V_n^{i+1} \right), \quad 0 \leq n \leq i, \quad i = M-1, \dots, 0. \quad (3)$$

The price at $t = 0$ is V_0^0 . For a European option this is all there is. For an American option one inserts an early-exercise comparison $V_n^i = \max(\Lambda(S_n^i), \text{continuation})$ at every node — this is precisely why the lattice is the simplest American pricer, a point developed in Lecture 9.

1.3 Derivation of the CRR parameters u, d, p

The parameters are fixed by *moment matching*: the one-period log-return of the tree is forced to share its mean and variance with the Black–Scholes log-return.

Introduce indicators $R_i \in \{0, 1\}$, where $R_i = 1$ if step i is an up-move. The R_i are i.i.d. Bernoulli(p). From (1), after n steps,

$$S(n\delta t) = S_0 u^{\sum_{i=1}^n R_i} d^{n - \sum_{i=1}^n R_i}, \quad \implies \quad \ln \frac{S(n\delta t)}{S_0} = n \ln d + \ln\left(\frac{u}{d}\right) \sum_{i=1}^n R_i.$$

Hence the log-price is an affine function of a Binomial(n, p) variable. Using $\mathbb{E}[R_i] = p$ and $\text{Var}(R_i) = p(1-p)$,

$$\mathbb{E} \left[\ln \frac{S(n\delta t)}{S_0} \right] = n(p \ln u + (1-p) \ln d), \quad (4)$$

$$\text{Var} \left[\ln \frac{S(n\delta t)}{S_0} \right] = n p(1-p) \left(\ln \frac{u}{d} \right)^2. \quad (5)$$

By the CLT the sum $\sum R_i$ is asymptotically Gaussian, so for large n the log-price is approximately normal — the correct distributional shape to match Black–Scholes. Under the risk-neutral measure the BS log-return over a horizon t has

$$\mathbb{E} \left[\ln \frac{S_t}{S_0} \right] = \left(r - \frac{1}{2} \sigma^2 \right) t, \quad \text{Var} \left[\ln \frac{S_t}{S_0} \right] = \sigma^2 t. \quad (6)$$

Matching (4)–(5) (per single step, $n = 1$, $t = \delta t$) against (6) gives the two moment equations

$$p \ln u + (1-p) \ln d = \left(r - \frac{1}{2} \sigma^2 \right) \delta t, \quad (7)$$

$$\ln\left(\frac{u}{d}\right) = \frac{\sigma \sqrt{\delta t}}{\sqrt{p(1-p)}}. \quad (8)$$

Equation (8) follows because $\text{Var} = \left(\ln \frac{u}{d} \right)^2 p(1-p) = \sigma^2 \delta t$, so $\ln \frac{u}{d} = \sigma \sqrt{\delta t} / \sqrt{p(1-p)}$.

Closing the system. We have two equations in the three unknowns (p, u, d) , so one degree of freedom remains. The lecture’s choice is the symmetric one,

$$p = \frac{1}{2}, \quad (9)$$

which gives $\sqrt{p(1-p)} = \frac{1}{2}$ and hence, from (8), $\ln \frac{u}{d} = 2\sigma\sqrt{\delta t}$, i.e.

$$\ln u - \ln d = 2\sigma\sqrt{\delta t}. \quad (10)$$

From (7) with $p = \frac{1}{2}$, $\frac{1}{2}(\ln u + \ln d) = (r - \frac{1}{2}\sigma^2)\delta t$, i.e.

$$\ln u + \ln d = 2\left(r - \frac{1}{2}\sigma^2\right)\delta t. \quad (11)$$

Adding and subtracting (10) and (11),

$$\boxed{u = \exp\left(\sigma\sqrt{\delta t} + \left(r - \frac{1}{2}\sigma^2\right)\delta t\right), \quad d = \exp\left(-\sigma\sqrt{\delta t} + \left(r - \frac{1}{2}\sigma^2\right)\delta t\right).} \quad (12)$$

This is the form used in the slides. (Note that with $p = \frac{1}{2}$ the tree is *not* the original Cox–Ross–Rubinstein parameterisation, in which one instead imposes $d = 1/u$ and solves for the implied p , yielding $u = e^{\sigma\sqrt{\delta t}}$, $d = e^{-\sigma\sqrt{\delta t}}$, $p = \frac{e^{r\delta t} - d}{u - d}$. Both match the first two moments to $O(\delta t)$; they differ only in the higher-order/free-parameter choice. The exam name “CRR” is used loosely here for the moment-matched binomial tree.)

Remark 1. *The matching is on the log of the price, not the price. Matching the variance of $\ln S$ to $\sigma^2\delta t$ is exactly the statement that the tree reproduces the diffusion coefficient of the GBM. The drift slot absorbs $(r - \frac{1}{2}\sigma^2)$, the Itô-corrected drift of $\ln S$, which is why $r - \frac{1}{2}\sigma^2$ (not r) appears in u, d .*

1.4 Convergence to Black–Scholes

For fixed T , as $M \rightarrow \infty$ (so $\delta t \rightarrow 0$) the per-step matching (7)–(8) plus the CLT force the terminal log-price distribution of the tree to converge to $\mathcal{N}((r - \frac{1}{2}\sigma^2)T, \sigma^2 T)$, which is the exact Black–Scholes terminal law. Since both prices are discounted expectations of the same payoff against converging laws, the binomial price converges to the BS price.

The convergence is *first order*, $|\text{error}| \leq K/M$ for some constant K , but it is *not monotone*: the error oscillates in sign as M increases (e.g. the slide’s European put: diff = $-0.0012, +0.0034, -0.0002$ at $M = 100, 200, 400$ against BS value 1.4728). The oscillation comes from how the discrete set of terminal nodes (1) straddles the strike K : the position of K relative to the lattice changes irregularly with M , so the payoff-region sampling error wobbles. This non-smoothness is why one cannot naively Richardson-extrapolate a raw binomial tree, and it motivates the smoother PDE/Fourier methods.

Pros / cons. Pros: trivial to describe and implement; readily extended to American and other early-exercise / path features for which no closed form exists. Cons: tied to GBM-type dynamics (more precisely, to models whose first moments of the log-return drive the price); irregular convergence.

2 Finite-difference methods for the heat equation

2.1 Model problem

Let $u(x, t)$ solve the (forward) heat equation on $x \in [0, L]$, $t \in [0, T]$,

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = a(t), \quad u(L, t) = b(t), \quad u(x, 0) = g(x). \quad (13)$$

A standard test case (used in the slides and in `finite-difference-heat_eqn.py`) is $L = \pi$, $g(x) = \sin x$, $a = b = 0$, with exact solution $u(x, t) = e^{-t} \sin x$.

2.2 Difference operators

For a smooth y sampled at spacing h , $y_m = y(mh)$, Taylor expansion gives the operators

$$\begin{aligned} \text{forward } \Delta y_m &= y_{m+1} - y_m &= hy' + \frac{1}{2}h^2y'' + \dots, & \text{backward } \nabla y_m &= y_m - y_{m-1} &= hy' - \frac{1}{2}h^2y'' + \dots \\ \text{central } \Delta_0 y_m &= \frac{1}{2}(y_{m+1} - y_{m-1}) &= hy' + \frac{1}{6}h^3y''' + \dots, & \text{2nd central } \delta^2 y_m &= y_{m+1} - 2y_m + y_{m-1} &= h^2y'' - \frac{1}{6}h^4y^{(4)} + \dots \end{aligned}$$

The two we need are the central second difference δ_x^2 (for u_{xx} , second-order accurate, $O(h^2)$) and a first time difference Δ_t (for u_t).

2.3 Grid

Discretise x into $N_x + 1$ points $x_j = jh$, $h = L/N_x$, and t into $N_t + 1$ points $t_i = ik$, $k = T/N_t$. Write $U_j^i \approx u(jh, ik)$. The *mesh ratio*

$$\nu := \frac{k}{h^2} \quad (14)$$

governs both accuracy and stability.

2.4 FTCS (explicit): Forward in Time, Central in Space

Forward difference in t , central second difference in x :

$$\frac{U_j^{i+1} - U_j^i}{k} = \frac{U_{j+1}^i - 2U_j^i + U_{j-1}^i}{h^2} \implies \boxed{U_j^{i+1} = \nu U_{j+1}^i + (1 - 2\nu) U_j^i + \nu U_{j-1}^i.} \quad (15)$$

The new value depends only on old values: the scheme is *explicit*. The stencil couples three space points at the old level and one at the new level. Collecting the $N_x - 1$ interior unknowns into $\mathbf{U}^i = (U_1^i, \dots, U_{N_x-1}^i)^\top$,

$$\mathbf{U}^{i+1} = F \mathbf{U}^i + \mathbf{p}^i, \quad F = \begin{pmatrix} 1 - 2\nu & \nu & & \\ \nu & 1 - 2\nu & \nu & \\ & \ddots & \ddots & \ddots \\ & & \nu & 1 - 2\nu \end{pmatrix}, \quad \mathbf{p}^i = \begin{pmatrix} \nu a(ik) \\ 0 \\ \vdots \\ 0 \\ \nu b(ik) \end{pmatrix}, \quad (16)$$

with $\mathbf{U}^0 = (g(h), \dots, g((N_x-1)h))^\top$. The boundary terms enter only the first and last components of $\mathbf{p}^i$ because the stencil for $j = 1$ and $j = N_x - 1$ reaches into the known boundary nodes U_0, U_{N_x} . (F is symmetric tridiagonal Toeplitz; in the code `finite-difference-heat_eqn.py` this is the explicit loop `u[n,i]=u_1[i]+nu*(u_1[i-1]-2u_1[i]+u_1[i+1]).`)

2.5 BTCS (implicit): Backward in Time, Central in Space

Backward difference in t (equivalently, evaluate the space operator at the *new* level):

$$\frac{U_j^{i+1} - U_j^i}{k} = \frac{U_{j+1}^{i+1} - 2U_j^{i+1} + U_{j-1}^{i+1}}{h^2} \implies U_j^i = -\nu U_{j+1}^{i+1} + (1 + 2\nu)U_j^{i+1} - \nu U_{j-1}^{i+1}. \quad (17)$$

Now the three unknowns sit at the new level: there is no explicit formula, one must solve a linear system each step,

$$B \mathbf{U}^{i+1} = \mathbf{U}^i + \mathbf{q}^i, \quad B = \begin{pmatrix} 1 + 2\nu & -\nu & & \\ -\nu & 1 + 2\nu & -\nu & \\ & \ddots & \ddots & \ddots \\ & & -\nu & 1 + 2\nu \end{pmatrix}, \quad \mathbf{q}^i = \begin{pmatrix} \nu a((i+1)k) \\ 0 \\ \vdots \\ 0 \\ \nu b((i+1)k) \end{pmatrix}. \quad (18)$$

B is tridiagonal and strictly diagonally dominant, so the Thomas algorithm solves each step in $O(N_x)$. The boundary data are taken at the new time $(i+1)k$ because the space stencil lives at the new level.

2.6 Crank–Nicolson

Centre the scheme at the half time level $i + \frac{1}{2}$ and average the space operator over the two surrounding levels (the time-averaging operator μ_t):

$$\frac{U_j^{i+1} - U_j^i}{k} = \frac{1}{h^2} \delta_x^2 \left(\frac{1}{2} U_j^{i+1} + \frac{1}{2} U_j^i \right). \quad (19)$$

Multiplying out the interior stencil,

$$-\frac{\nu}{2} U_{j-1}^{i+1} + (1 + \nu) U_j^{i+1} - \frac{\nu}{2} U_{j+1}^{i+1} = \frac{\nu}{2} U_{j-1}^i + (1 - \nu) U_j^i + \frac{\nu}{2} U_{j+1}^i, \quad (20)$$

i.e. in matrix form $B \mathbf{U}^{i+1} = F \mathbf{U}^i + \mathbf{r}^i$ with

$$B = \text{tridiag}\left(-\frac{\nu}{2}, 1 + \nu, -\frac{\nu}{2}\right), \quad F = \text{tridiag}\left(\frac{\nu}{2}, 1 - \nu, \frac{\nu}{2}\right), \quad \mathbf{r}_1^i = \frac{\nu}{2} (a(ik) + a((i+1)k)), \text{ etc.}$$

CN is implicit (one tridiagonal solve per step) but is the average of FTCS and BTCS, and as we see below it gains an order in time.

2.7 Local truncation error

Substitute the exact solution into the scheme and Taylor-expand; the residual measures consistency. Using

$$\frac{u(x, t+k) - u(x, t)}{k} = u_t + \frac{1}{2} k u_{tt} + O(k^2), \quad \frac{u(x+h, t) - 2u(x, t) + u(x-h, t))}{h^2} = u_{xx} + \frac{1}{12} h^2 u_{xxxx} + O(h^4),$$

and the PDE $u_t = u_{xx}$ (which cancels the leading terms), the FTCS residual is

$$R_j^i = \frac{1}{2} k u_{tt} - \frac{1}{12} h^2 u_{xxxx} + O(k^2) + O(h^4) \implies \text{FTCS: } O(k) \text{ in time, } O(h^2) \text{ in space.} \quad (21)$$

For BTCS the time term flips sign, $R_j^i = -\frac{1}{2} k u_{tt} - \frac{1}{12} h^2 u_{xxxx} + \dots$, again $O(k) + O(h^2)$. For Crank–Nicolson the symmetric (midpoint) time-centring kills the $O(k)$ term, leaving

$$R_j^i = O(k^2) + O(h^2) \implies \text{CN: } O(k^2) \text{ in time, } O(h^2) \text{ in space.}$$

This second-order-in-time accuracy at the same per-step cost as BTCS is the practical reason CN is preferred.

2.8 Von Neumann stability analysis

Consistency alone is not enough; the scheme must be stable. *Lax equivalence*: for a consistent linear scheme, convergence $\iff$ stability. Von Neumann tests stability by Fourier mode analysis. Because the schemes are linear and shift-invariant in space, the error $e_j^i = U_j^i - u_j^i$ decomposes into independent Fourier modes

$$e_j^i = \sum_{\beta} \hat{e}_{\beta}^i e^{i\beta j h},$$

and each mode is multiplied per step by an *amplification factor* $\xi(\beta)$. Substituting the single mode $U_j^i = \xi^i e^{i\beta j h}$ into the scheme, the scheme is stable iff

$$|\xi(\beta)| \leq 1 \quad \text{for all } \beta h \in [-\pi, \pi]. \quad (22)$$

FTCS. Insert $U_j^i = \xi^i e^{i\beta j h}$ into (15) and divide by $\xi^i e^{i\beta j h}$:

$$\xi = \nu e^{i\beta h} + (1 - 2\nu) + \nu e^{-i\beta h} = 1 + \nu(e^{i\beta h} - 2 + e^{-i\beta h}).$$

Using $e^{i\theta} - 2 + e^{-i\theta} = (e^{i\theta/2} - e^{-i\theta/2})^2 = (2i \sin \frac{\theta}{2})^2 = -4 \sin^2 \frac{\theta}{2}$ with $\theta = \beta h$,

$$\xi(\beta) = 1 - 4\nu \sin^2\left(\frac{\beta h}{2}\right). \quad (23)$$

This ξ is real, with $\xi \leq 1$ automatically. The binding constraint is $\xi \geq -1$:

$$1 - 4\nu \sin^2\left(\frac{\beta h}{2}\right) \geq -1 \iff 4\nu \sin^2\left(\frac{\beta h}{2}\right) \leq 2 \iff \nu \sin^2\left(\frac{\beta h}{2}\right) \leq \frac{1}{2}.$$

The worst mode is $\sin^2 = 1$ (the highest frequency, $\beta h = \pi$), giving the **CFL-type stability restriction**

$$\boxed{\nu = \frac{k}{h^2} \leq \frac{1}{2} \iff k \leq \frac{h^2}{2}.} \quad (24)$$

This is exactly the slide's statement "FTCS is stable when $\nu \leq 0.5$ " and is visible in the code: $\nu \approx 0.30$ gives a clean surface (error $\sim 10^{-3}$), $\nu \approx 0.63$ blows up (the highest-frequency saw-tooth mode is amplified, $|\xi| > 1$). The cost of explicitness is the quadratic coupling $k \leq h^2/2$: halving h forces a fourfold cut in k .

BTCS. Insert the mode into (17): $\xi - 1 = \nu \xi (e^{i\beta h} - 2 + e^{-i\beta h}) = -4\nu \xi \sin^2 \frac{\beta h}{2}$, hence

$$\xi(\beta) = \frac{1}{1 + 4\nu \sin^2\left(\frac{\beta h}{2}\right)} \in (0, 1] \quad \text{for all } \nu > 0. \quad (25)$$

Thus BTCS is **unconditionally stable**: no restriction on k . Crank–Nicolson, with $\xi = \frac{1 - 2\nu \sin^2(\beta h/2)}{1 + 2\nu \sin^2(\beta h/2)}$, has $|\xi| \leq 1$ for all $\nu > 0$ and is also unconditionally stable (and, since $\xi \rightarrow -1$ for large ν on high modes, it is only damped, not strongly dissipative — the reason CN can show mild oscillations on non-smooth initial data such as an option payoff kink).

3 Solving the Black–Scholes PDE

3.1 Reverse-time form

The Black–Scholes PDE carries a *final*-time condition (the payoff at $t = T$). Substituting $\tau = T - t$ turns it into an initial-value problem (now ∂_τ replaces $-\partial_t$):

$$\frac{\partial V}{\partial \tau} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rS \frac{\partial V}{\partial S} + rV = 0. \quad (26)$$

The domain $S \in [0, \infty)$ is truncated at a large L . For a European put (E the strike),

$$V(S, 0) = \max(E - S, 0), \quad V(0, \tau) = Ee^{-r\tau}, \quad V(L, \tau) = 0,$$

and for a call $V(S, 0) = \max(S - E, 0)$, $V(0, \tau) = 0$, $V(L, \tau) = L - Ee^{-r\tau}$.

3.2 FTCS for Black–Scholes

On the uniform S -grid $S_j = jh$ the variable coefficients $S^2 \rightarrow (jh)^2$ and $S \rightarrow (jh)$ enter explicitly. Central differences for both space derivatives:

$$\frac{V_j^{i+1} - V_j^i}{k} - \frac{1}{2}\sigma^2(jh)^2 \frac{V_{j+1}^i - 2V_j^i + V_{j-1}^i}{h^2} - r(jh) \frac{V_{j+1}^i - V_{j-1}^i}{2h} + rV_j^i = 0.$$

The h^2 in the diffusion term cancels against $(jh)^2/h^2 = j^2$, and h cancels in the convection term, leaving the explicit update

$$\boxed{V_j^{i+1} = (1 - rk)V_j^i + \frac{1}{2}k\sigma^2 j^2 (V_{j+1}^i - 2V_j^i + V_{j-1}^i) + \frac{1}{2}krj(V_{j+1}^i - V_{j-1}^i).} \quad (27)$$

In matrix form $\mathbf{V}^{i+1} = F\mathbf{V}^i + \mathbf{p}^i$ with $F = (1 - rk)I + \frac{1}{2}k\sigma^2 D_2 T_2 + \frac{1}{2}kr D_1 T_1$, where

$$D_1 = \text{diag}(1, 2, \dots, N_x - 1), \quad D_2 = \text{diag}(1^2, 2^2, \dots, (N_x - 1)^2),$$

$T_2 = \text{tridiag}(1, -2, 1)$ is the second-difference matrix and $T_1 = \text{tridiag}(-1, 0, 1)$ the central-first-difference matrix. The diagonal weights j and j^2 are the discrete images of the S and S^2 coefficients — this is exactly the structure built in `ftcs_bs_put.py` (there written on the log-grid; see Remark below). BTCS evaluates the same spatial operator at level $i+1$, giving $B\mathbf{V}^{i+1} = \mathbf{V}^i + \mathbf{q}^i$ with $B = (1 + rk)I - \frac{1}{2}k\sigma^2 D_2 T_2 - \frac{1}{2}kr D_1 T_1$; Crank–Nicolson is the average,

$$\left(\frac{1}{2}I + B\right)\mathbf{V}^{i+1} = \left(\frac{1}{2}I + F\right)\mathbf{V}^i + \frac{1}{2}(\mathbf{p}^i + \mathbf{q}^i).$$

Remark 2 (The S -grid vs. the $\ln S$ -grid, and the Assignment-2 truncation issue). *On the raw S -grid the diffusion weight $\frac{1}{2}k\sigma^2 j^2$ grows like j^2 . For explicit FTCS the von Neumann condition therefore reads, mode by mode, $\sigma^2 j^2 k/1 \lesssim 1$ at the largest active $j = N_x - 1$, i.e. stability degrades quadratically near the upper boundary $S = L$. The companion file `ftcs_bs_put.py` sidesteps this by working in $x = \ln S$ on $[-L/2, L/2]$, where the PDE has constant coefficients (Section 4) and the stencil reduces to a constant-coefficient advection–diffusion stencil. This is the same lesson that bit me in Assignment 2 (COS): a numerical method built on a finite spatial window is only as good as the window. There the COS truncation range $[a, b] = [c_1 \pm L\sqrt{c_2 + \sqrt{c_4}}]$ collapsed at short maturity because the variance cumulant $c_2 \propto T$ becomes tiny, so $[a, b]$ became too narrow and the density support was clipped. The FD analogue is exactly choosing L (and the grid) so the truncated domain still contains essentially all of the relevant payoff/density mass; the difference is only that FD truncates the state space directly whereas COS truncates the Fourier-cosine integration range.*

4 Reduction of the BS PDE to the heat equation

This is the transformation that links Part 3 to Part 2 and underlies the next section.

Step 1: log-price. Set $X = \ln S$. Then $\partial_S = \frac{1}{S}\partial_X$ and $\partial_{SS} = \frac{1}{S^2}(\partial_{XX} - \partial_X)$, so $S^2\partial_{SS} = \partial_{XX} - \partial_X$ and $S\partial_S = \partial_X$. Equation (26) becomes a *constant-coefficient* equation,

$$\frac{\partial V}{\partial \tau} - \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial X^2} - \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial V}{\partial X} + rV = 0. \quad (28)$$

Step 2: remove the reaction term. Put $V = e^{-r\tau}W$ (equivalently the slide’s $V = e^{rt}W$). Then $\partial_\tau V = e^{-r\tau}(\partial_\tau W - rW)$, and the rV and $-rW$ terms cancel:

$$\frac{\partial W}{\partial \tau} - \frac{1}{2}\sigma^2 \frac{\partial^2 W}{\partial X^2} - \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial W}{\partial X} = 0. \quad (29)$$

This is a pure advection–diffusion equation with constant drift $r - \frac{1}{2}\sigma^2$ and diffusivity $\frac{1}{2}\sigma^2$.

Step 3 (to a pure heat equation). A further change to a moving frame, $\xi = X + (r - \frac{1}{2}\sigma^2)\tau$, removes the advection term and a rescaling $\tau \mapsto \frac{1}{2}\sigma^2\tau$ normalises the diffusivity to 1, yielding $\partial_\tau W = \partial_{\xi\xi} W$, the model heat equation (13). *(The slides stop at (29); Step 3 is standard and is included here for completeness — it is what justifies importing the heat-equation FD theory wholesale.)*

5 Binomial method as an explicit finite-difference scheme

The closing result of the lecture: the binomial recursion (3) is a finite-difference scheme applied to (29), with the grid and time step chosen in a special way.

Start from the constant-coefficient log-equation (29) and apply BTCS in τ with central space differences (here I follow the slide's discretisation, which evaluates the space operator at the new level):

$$\frac{W_j^{i+1} - W_j^i}{k} + \frac{\sigma^2}{2} \frac{W_{j+1}^{i+1} - 2W_j^{i+1} + W_{j-1}^{i+1}}{h^2} + \left(r - \frac{1}{2}\sigma^2\right) \frac{W_{j+1}^{i+1} - W_{j-1}^{i+1}}{2h} = 0. \quad (30)$$

(The signs follow from moving the diffusion/convection terms of (29) to evaluate the recursion backward in τ .) The key trick is the *coupling of the two grid spacings*:

$$h^2 = \sigma^2 k. \quad (31)$$

With this choice the coefficient of the centre value W_j^{i+1} collapses. The $\frac{\sigma^2}{2} \cdot \frac{-2}{h^2} W_j^{i+1} = -\frac{\sigma^2}{h^2} W_j^{i+1}$ from the diffusion term is, by (31), $= -\frac{1}{k} W_j^{i+1}$, which exactly cancels the $\frac{1}{k} W_j^{i+1}$ produced by the time difference. The centre node disappears and we are left with a two-point (up/down) recursion linking level i to level $i+1$:

$$W_j^i = p^* W_{j+1}^{i+1} + (1 - p^*) W_{j-1}^{i+1}, \quad (32)$$

a convex combination of the up-neighbour $j+1$ and the down-neighbour $j-1$ — structurally identical to the binomial backward step (3). Collecting the coefficients of W_{j+1}^{i+1} (diffusion $\frac{\sigma^2}{2h^2} = \frac{1}{2k}$ plus convection $+\frac{r-\sigma^2/2}{2h}$) and multiplying through by k , the up-weight is

$$p^* = \frac{1}{2} \left(1 + \frac{(r - \frac{1}{2}\sigma^2)k}{h} \right) = \frac{1}{2} \left(1 + \frac{(r - \frac{1}{2}\sigma^2)\sqrt{k}}{\sigma} \right), \quad (33)$$

using $h = \sigma\sqrt{k}$ from (31). Undoing the discounting substitution $V = e^{-r\tau}W$ (one step of τ multiplies by e^{-rk}) restores the discount factor and gives

$$\boxed{V_j^i = e^{-rk} (p^* V_{j+1}^{i+1} + (1 - p^*) V_{j-1}^{i+1})}. \quad (34)$$

This is precisely the risk-neutral binomial backward induction (3), with up-probability p^* and an up/down log-spacing of $\pm h = \pm\sigma\sqrt{k}$ (i.e. $u/d = e^{\pm\sigma\sqrt{k}}$ in the moving log-frame). The binomial method is therefore an explicit FD scheme on the transformed heat equation, “customised” (via $h^2 = \sigma^2 k$) so that the centre node drops out and only the two outer branches survive — and tuned to deliver the single time-zero value V_0^0 rather than the whole grid. A general FD scheme instead returns the option value at *every* grid point $\{jh, ik\}$, hence prices for all initial spots $S_0 \in \{0, h, 2h, \dots, L\}$ at once.

Remark 3 (Honesty: a typo in the slide's p^*). *Slide 57 prints $p^* = \frac{1}{2}(1 + \sqrt{k}/\sigma \cdot r - \sigma^2)$, which is dimensionally inconsistent (the $-\sigma^2$ is not divided by anything and a factor is missing). The correct algebra, carried out above, gives (33), $p^* = \frac{1}{2}(1 + \frac{(r-\sigma^2/2)\sqrt{k}}{\sigma})$, which is dimensionless and reduces to $\frac{1}{2}$ as $k \rightarrow 0$ — consistent with the symmetric $p = \frac{1}{2}$ choice (9) of the tree. I treat the slide formula as a transcription error.*

Honest gaps / things the slides leave implicit

- The slides assert first-order non-monotone binomial convergence and quote Higham’s numbers but give no proof of the K/M bound; the oscillation mechanism (strike position relative to lattice) is my added explanation, standard but not on the slides.
- Step 3 of the heat-equation reduction (moving frame + time rescale to reach $\partial_\tau W = \partial_{\xi\xi} W$) is mine; the deck stops at the constant-coefficient advection–diffusion form (29).
- The p^* formula on Slide 57 is garbled; (33) is the corrected version.
- The BS-PDE FTCS stability condition is only stated qualitatively in the deck. A full mode analysis on the variable-coefficient S -grid is more delicate than the constant-coefficient $\nu \leq \frac{1}{2}$; the practical rule is that the largest diffusion weight $\sigma^2(N_x - 1)^2 k \lesssim 1$ controls it. I flag this as a likely deeper-probe question.
- “CRR” nomenclature: the deck’s $p = \frac{1}{2}$ tree is moment-matched but is not the classical $d = 1/u$ CRR tree; I noted both parameterisations.